Topic 8 — Minerals, Energy & Industry

Sources baked in: NCERT Class 10–12 minerals/energy/industry, Drishti / PW heads, **Ghatnachakra** Mineral Resources (CA-264+ rock systems → metals → coal/oil → power → industry), UPPCS Prelims 2018–2025 **Weight:** ★★★★ — mine–state triples, Dharwar vs Gondwana, coalfield traps, nuclear/hydel, refineries, corridors **Last verified:** August 2026 **Current Affairs:** Critical minerals (Li Reasi J&K); **National Critical Mineral Mission**; energy mix / nuclear+geothermal ≠ solar; DFC/DMIC as living schemes

Current Affairs (this topic)

Year	Fact	Why asked	Source
2023	Li inferred resources **Reasi, J&K**; India import-dependent for **Li, Ni, Co**	Frozen 2023 Q77	MoM / PYQ
2023	Union list (संघ सूची) of **24 critical minerals** (Li, Ni, Co, REE, graphite, etc.)	Scheme/list trap	Ministry of Mines
2023–26	**Critical Minerals Mission** + auction of critical-mineral blocks	Living CA	PIB / MoM
2015 / 2025	**ISA**: India+France, HQ **Gurugram**; **not 38 members**; **OSOWOG** under ISA	2019 Q63 + 2025 Q109	ISA / MEA
Jan 2026	US announced withdrawal from ISA; ISA continues with **120+** members	Member-count trap	MEA / ET
2018	Major solar plant **Mirzapur (मिर्ज़ापुर) (UP)** with **France**	UP energy PYQ	2020 Q46
DFC	**Western** Dadri (ददरी)–JNPT; **Eastern** Ludhiana–Dankuni — phased commissioning	Corridor syllabus	DFCCIL
Nuclear	Kakrapar / Kudankulam expansion; **Narora = only UP** plant	Plant–state	NPCIL

UKPCS Prelims 2025

Q. UKPCS Prelims 2025, Q100

Which is the largest centre of the petrochemical industry in India?

A. Trombay B. Jamnagar C. Digboi D. Vadodara

Show answer

Logic: Digboi is oldest, not largest. **Ans: B.** Jamnagar hosts the largest refining–petrochemical complex. Digboi is oldest but small.

N.0 Framework (map first)

Mineral is a naturally occurring inorganic substance with a definite chemical composition. **Ore** is the rock from which a metal can be extracted at a profit.

- Most metallic minerals sit in the **peninsular plateau**, because old crystalline and metamorphic rocks there hold the ores.
- Most major mineral resources of India lie **east of a line joining Mangaluru (Mangalore) and Kanpur**.
- About **97% of India's coal** sits in the valleys of the **Damodar, Son, Mahanadi and Godavari**.
- Petroleum and natural gas sit in **sedimentary basins**, not in the old crystalline shield of Chota Nagpur.
- The vast **northern alluvial (जलोढ़) plains are almost devoid of economic minerals**.

A deposit becomes a **mine** only when grade, ease of extraction, and distance to market make working it profitable.

Rock systems (Ghatnachakra / NCERT map)

System	Character	Main minerals
Dharwar	Oldest metamorphosed belts; first studied in Karnataka	**Storehouse of metallic minerals** — iron, manganese, gold, copper; most mineralised system in teaching keys
Gondwana	Palaeozoic–Mesozoic sediments in eastern/central valleys	About **95–99%** of India's **coal** (Damodar, Son, Mahanadi, Godavari, etc.)
Vindhyan	Sedimentary (limestone, sandstone, shale)	**Limestone**, sand, china clay — cement base; **not** the metallic king
Cuddapah / Tertiary	Younger / basin rocks	Tertiary coal NE; petroleum in sedimentary basins (Gujarat, Assam, offshore)

- Peninsular India is the **heart-place** of Indian minerals. **Chota Nagpur** is the richest plateau belt (often "Ruhr of India").
- Alluvial north India is almost empty of economic ores. Rajasthan's Aravalli (अरावली) rock frame holds many **non-ferrous** minerals (Cu, Zn–Pb, etc.).
- In Ministry of Mines **2021–22** data (metallic and non-metallic value, excluding fuel, atomic and minor minerals), the leading order is **Odisha**, then **Chhattisgarh**, then **Karnataka**, then **Rajasthan** (थार), then **Jharkhand**.
- India produces about **95** minerals as listed in the Mines Annual Report (fuel / metallic / non-metallic / atomic / minor counts vary by year).

How minerals occur

NCERT teaches **five** modes. Vein is the **smaller** crack fill; lode is the **larger** crack fill.

Mode	How it forms	Examples
Vein / lode	Molten / gaseous minerals rise into cracks of **igneous and metamorphic** rock, then cool	Tin, copper, zinc, lead
Bed / layer	Deposition in **sedimentary** strata under heat and pressure	Coal, iron ore, gypsum, potash, common salt (लवणाध्यक्ष)
Residual	Surface rock weathers; soluble parts wash away; a concentrated cap remains	**Bauxite** on laterite
Placer / alluvial	Hard grains collect in valley-floor / hill-base sands (not corroded by water)	Gold, silver, tin, platinum, **monazite**
Ocean water / ocean bed	Dissolved salts, and nodules on the sea floor	Common salt, magnesium, bromine; **manganese nodules**

Mining is **opencast** when the ore is near the surface, and **underground** when it is deep. Opencast is cheaper but scars land; underground is costlier and more hazardous. Narrow family tunnels in Meghalaya coal are called **rat-hole** mining (NGT has restricted the practice).

Classification

Class	Meaning	Examples
Ferrous metallic	Contain iron / feed steel	Iron ore, manganese, chromite, nickel, cobalt
Non-ferrous metallic	Metals without iron as the main metal	Bauxite (aluminium), copper, zinc, lead, tin
Precious metals	High-value non-ferrous subset	Gold, silver, platinum
Non-metallic (inorganic)	Not used mainly as metal	Mica, limestone, gypsum, graphite, diamond, salt, magnesite
Energy / mineral fuels	Organic fossil fuels + nuclear fuels	Coal, petroleum, natural gas, uranium, thorium

- India is **rich in ferrous minerals**, but **poor in non-ferrous metals except bauxite**.
- Ferrous minerals make up about **three-fourths** of the value of metallic mineral production.
- Minerals take geological ages to form, so they are **exhaustible / non-renewable** at human timescales.

Mineral belts of India

The mineral belts of India form one map. The north-eastern plateau is richest; the other belts add specific minerals.

Belt	Cover	Character
North-Eastern Plateau (Chota Nagpur)	Jharkhand, Odisha, West Bengal, parts of Chhattisgarh	**Richest** — iron, coal, manganese, bauxite, mica, copper, chromite; often called **India's Ruhr**
Central belt	Chhattisgarh, Madhya Pradesh, Telangana, Andhra Pradesh, Maharashtra	Manganese, bauxite, limestone, coal, iron, mica, graphite, gems
South-Western Plateau	Karnataka, Goa, adjoining Tamil Nadu–Kerala uplands	Ferrous metals, bauxite, limestone; **almost no Gondwana coal**; **Neyveli lignite** in Tamil Nadu; Kerala **monazite** sands; Goa iron
North-Western (Aravalli)	Rajasthan–Gujarat	Copper, zinc–lead, mica, building stone, gypsum, **Fuller's earth**, salt; Gujarat petroleum
Himalayan / Assam–Arakan	North-East and Himalayan foothills	Copper–lead–zinc pockets; **Tertiary coal**; Assam oil

Master map — mineral by state / centre (revise daily)

Mineral	Main state / centre	One-line note
Iron ore	**Odisha**, Chhattisgarh, Karnataka, Jharkhand	Bailadila (CG); Noamundi (JH); Kudremukh / Ballari (KA)
Manganese	**Odisha**, Maharashtra, Madhya Pradesh	Balaghat (MP); Nagpur–Bhandara (MH)
Chromite	**Odisha (Sukinda)**	Stainless steel / ferrochrome
Bauxite	**Odisha (Panchpatmali)**	Residual laterite ore of aluminium
Copper	**Khetri (RJ)**; Malanjkhand (MP); Singhbhum (JH)	India still imports copper
Zinc–lead	**Zawar**; **Rampura-Agucha (Bhilwara)** — RJ	Rajasthan near-monopoly
Gold	**Hutti** (working); **Kolar** (historic (ऐतिहासिक)) — KA	Tiny mine output vs jewellery
Mica	**Koderma–Gaya–Hazaribagh**; Nellore; Ajmer (अजमेर)–Bhilwara	Dielectric sheets
Diamond	**Panna (MP)**	India's diamond centre
Coal (Gondwana)	Damodar–Son–Mahanadi–Godavari valleys	~**97%** of Indian coal
Lignite	**Neyveli (TN)**	Brown coal; pit-head power
Petroleum	Assam; Mumbai High; Gujarat; Barmer; KG / Cauvery	Sedimentary basins only
Uranium	**Jaduguda (JH)**	Also Tummalapalle (AP); Domiasiat (ML)

Mineral	Main state / centre	One-line note
Thorium / monazite	**Kerala–Tamil Nadu** beach sands	Also rare earths

Conservation

- Recycle metals and use **scrap** in steel and aluminium.
- Use **substitutes** where a scarce metal is not essential.
- Improve technology so **low-grade ores** become usable.
- Control mine safety, land restoration, and pollution, because unmanaged mining is a high-risk industry.

N.1 Metallic Minerals



Iron

Iron is the backbone of heavy industry. Steel needs iron ore plus **coking coal** and a **flux** such as limestone or dolomite.

Ore	Iron content	Trait
Magnetite (Fe_3O_4)	About **70%**	Finest / richest ore; magnetic; electrical industry
Hematite (Fe_2O_3)	About **50–60%**	Most used industrial ore by quantity

Ore	Iron content	Trait
Limonite	About **40–60%** (lower / variable)	Hydrated iron oxide; lesser industrial weight
Siderite	About **40–50%** (lower)	Iron carbonate ore; lesser industrial weight

- About **95%** of India's iron-ore reserves sit in Odisha, Jharkhand, Chhattisgarh, Karnataka, Goa, Telangana, Andhra Pradesh and Tamil Nadu.
- Iron mines of the north-eastern plateau often sit **near coalfields**, which helps steel plants.
- As per the Ministry of Mines Annual Report **2021–22**, the leading iron-ore producers are **Odisha and Chhattisgarh** (then Karnataka, Jharkhand).
- About **79%** of hematite resources sit in the **eastern** sector.
- About **93%** of magnetite resources sit in the **southern** sector (Karnataka leads magnetite resources).
- **Hematite** has the largest iron-ore **reserve** volume in India among ore types.
- **Nathra-ki-Pal / Thoor–Hunder** (Udaipur (उदयपुर)) is the Rajasthan iron centre.
- **Chiria** (West Singhbhum) is an iron-mining centre.
- **Ballari** and **Chitradurga** belong to the Karnataka **iron** belt.
- **Mount Newman** in Australia is famous for **iron ore**.

Four iron-ore belts

Belt	Note
Odisha–Jharkhand	Mayurbhanj (मयूरभंज)–Keonjhar–Sundargarh hematite; Gua–Noamundi in Singhbhum
Durg–Bastar–Chandrapur	Bailadila high-grade hematite in Chhattisgarh; Dalli–Rajhara; export via Visakhapatnam
Ballari–Chitradurga–Chikkamagaluru–Tumakuru	Karnataka magnetite–hematite; Kudremukh slurry to Mangaluru
Maharashtra–Goa	Lower grade; export via **Mormugao**
Mine / belt	State
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Gurumahisani / Badampahar / Sulaipat	**Odisha**
Bonai / Kendujhar / Sundargarh	Odisha
Bailadila	**Chhattisgarh**
Dalli–Rajhara	Chhattisgarh
Noamundi / Gua / Kiriburu	**Jharkhand**
Kudremukh	**Karnataka**
Donimalai / Ballari–Hospet	Karnataka
Ratnagiri belt	Maharashtra

Manganese

Manganese is a **ferrous** mineral. It hardens steel and is also used in bleaching powder, insecticides, paints and dry batteries.

Point	Teaching line
Steel need	About **10 kg of manganese** per tonne of steel
Rock link	Mainly associated with **Dharwar** rocks
Leading states	**Odisha, Maharashtra, Madhya Pradesh** (Karnataka is another classic belt)
Madhya Pradesh belt	**Balaghat**
Maharashtra belt	**Nagpur–Bhandara**
Odisha belt	With the iron belt of **Keonjhar–Sundargarh**
World centre	**Postmasburg** (South Africa)

Chromite

Chromite is the ore of **chromium**. It feeds **stainless steel** and **ferrochrome**.

Point	Teaching line
Main Indian belt	**Sukinda in Odisha** (India's leading chromite belt)
Use	**Stainless steel** and **ferrochrome**
Belt family	Eastern plateau ferrous belt with Odisha iron–manganese country

Bauxite and aluminium

Bauxite is the ore of **aluminium**. It commonly forms as a **residual** weathering cap on **laterite** in hot-wet plateaus.

Point	Teaching line
Leading state	**Odisha**
Leading mine	**Panchpatmali** (Koraput, Odisha)
Other belts	Gujarat, Jharkhand, Chhattisgarh, Maharashtra
Older city tag	**Katni** (Madhya Pradesh)
Smelting logic	Aluminium smelting is **power-hungry**, so plants sit near cheap electricity as well as near ore
Company (कंपनी)	Plant
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BALCO	**Korba**
HINDALCO	**Renukoot**

Point	Teaching line
MALCO	**Mettur**
INDAL	**Hirakud** (हीराकुंड)
NALCO	Damanjodi (alumina) / Angul (smelter)

Copper

Copper is valued for **electrical conductivity** (wires, alloys, electronics). India is **not copper-rich** and still imports a large share of its need.

Centre	State	Note
Khetri (Khetri–Singhana belt)	**Rajasthan**	Classic textbook copper centre
Malanjkhand	**Madhya Pradesh**	Large **open-pit** copper mine
Singhbhum (Mosabani–Ghatsila)	**Jharkhand**	Eastern copper belt
World lead	**Chile** (northern **Andes porphyry**)	Leading producer in the usual paper set

Gold

Centre	State	Note
Kolar	Karnataka	Historic goldfield; now largely closed
Hutti	Karnataka	Main working goldfield
Ramagiri	Andhra Pradesh	Smaller goldfield

- India’s mine output is tiny beside jewellery demand.

Tin

Point	Teaching line
Ore	**Cassiterite**
Only producing state (recent IMYB)	**Chhattisgarh** (Dantewada belt)
Reserve tables	Haryana also appears in resource share
World centres	**Kinta Valley (Malaysia)**; **Bangka (Indonesia)**

- India has very little tin overall.

Zinc, lead and silver

Centre	State	Metal
Zawar (near Udaipur)	Rajasthan	Historic **zinc–lead** (silver often a by-product)
Rampura-Agucha (Bhilwara)	Rajasthan	Asia's richest zinc–lead reserve belt; Hindustan Zinc link

- Rajasthan has a near **monopoly** of India's zinc–lead ore resources and production.
- Silver usually occurs **with lead–zinc**; Rajasthan leads silver output in recent Indian Minerals Yearbook data.

Nickel and cobalt

- India is **import-dependent** for nickel and cobalt.
- They occur in laterite and some ultramafic belts, but India has no Khetri-class nickel–cobalt mines.

N.2 Non-metallic and atomic minerals

Mica

Mica splits into thin flexible sheets. It is a **dielectric / electrical insulator** used in capacitors, electrical equipment and heat-resistant sheets.

Belt	State / region	Note
Koderma–Gaya–Hazaribagh–Munger	Jharkhand–Bihar	Classic belt; **Kodarma** often called the "capital of mica"
Nellore	Andhra Pradesh	Southern mica centre; AP leads **resource** share in IMYB
Ajmer–Bhilwara	Rajasthan	Rajasthan leads recent **production**; Bhilwara is also zinc country

- Mica was notified as a **minor mineral** (2015).
- China leads world mica production in recent World Mineral Production data; older Indian notes stressed India's sheet-mica lead.
- **Makrana** (Nagaur, Rajasthan) supplies India's best marble (Taj Mahal (ताज महल) / Victoria Memorial); marble is metamorphic (recrystallised limestone).

Limestone, dolomite, gypsum, cement rocks

Rock / mineral	Use	Where (teaching centres)
Limestone	Cement; bulky → plant near quarry	UP: **Chunar–Churk–Dalla**; wide Vindhyan / peninsula belts

Rock / mineral	Use	Where (teaching centres)
Dolomite	Steel **flux**; refractory	UP: **Sonbhadra–Mirzapur**; Odisha and other plateau belts
Gypsum	Cement; reclaim **usar / alkaline** soils	Rajasthan; western Uttar Pradesh

Graphite and diamond

Mineral	Where	Use / note
Graphite	Odisha, Jharkhand	Crucibles, pencils, electrodes; carbon mineral
Diamond	**Panna (Madhya Pradesh)**	India's main diamond centre

- Graphite and diamond are both carbon, but they are different minerals.

Salt, magnesite, silica, Fuller's earth

Mineral	Chemistry / use	Where
Common salt	Food; chemicals; from sea / lakes	**Sambhar Lake (RJ)** inland; Gujarat coastal solar salt
Magnesite	$MgCO_3$; **refractory** bricks	Tamil Nadu; Uttarakhand (उत्तराखंड)
Silica sand	Glass sand	**Shankargarh (Prayagraj (प्रयागराज))** → **Firozabad** glass
Fuller's earth	Clay used to clean oils and grease	North-western **Aravalli** (अरावली) belt, Rajasthan

Other industrial minerals

Mineral	Use	Where
Kyanite / sillimanite	Refractories	Eastern plateau (Jharkhand–Odisha)
Asbestos	Older industrial fibre	Rajasthan; Andhra belts
Phosphorite	Fertilizer phosphate	**Jhamarkotra (Rajasthan)**; Bundelkhand / Lalitpur

Atomic minerals and critical minerals

Mineral	Main centre	Note
Uranium	**Jaduguda (Jharkhand)**	Singhbhum shear; also Bhatin–Narwapahar–Turamdih
Uranium (other)	**Tummalapalle (AP)**; **Domiasiat / Mawthabah (Meghalaya)**	Extra centres

Mineral	Main centre	Note
Thorium / monazite	**Kerala–Tamil Nadu** beach sands	Also Andhra–Odisha coasts; monazite carries **rare earths**
Lithium	**Reasi, Jammu & Kashmir**	2023 inferred-resource find; India still imports Li
Nickel / cobalt	Import-dependent	Laterite / ultramafic traces; no Khetri-class mines

- Shankargarh silica is **glass sand**, not an atomic mineral.
- The Union **critical minerals** list includes lithium, nickel, cobalt, rare earths, graphite and others.
- Nuclear fuel is **not** solar-derived energy.

Inline PYQ — UPPCS Prelims 2025, Q108

Match List-I with List-II and choose the correct answer using the code given below.

List-I (Iron Ore Mine)	List-II (State)
A. Gurumahisani	1. Jharkhand
B. Bailadila	2. Karnataka
C. Noamundi	3. Odisha
D. Kudremukh	4. Chhattisgarh

> Row order is **not** the answer — use the option codes.

A. 3 4 2 1

B. 4 3 1 2

C. 4 3 2 1

D. 3 4 1 2

Show answer

Ans: D — Gurumahisani–Odisha; Bailadila–Chhattisgarh; Noamundi–Jharkhand; Kudremukh–Karnataka.

N.3 Coal

Coal is a sedimentary **fossil fuel** formed from buried plant matter. It is **non-renewable**. Rank rises as **carbon rises** and **moisture falls**.

Two geological ages

Age	Share / quality	Where found
Gondwana coal	About **97–98%** of India's coal; mostly **bituminous**; usually low sulphur, high ash	River valleys of the peninsula: **Damodar, Son, Mahanadi, Godavari** (and Wardha)
Tertiary coal	Small share; younger; often **high sulphur**	**Assam, Meghalaya, Nagaland, Arunachal Pradesh**; also Kalakot (J&K) in older NCERT lists

Types of coal (carbon % and Indian homes)

Soft grades harden toward anthracite in this order: **peat**, then **lignite**, then **bituminous**, then **anthracite**.

Type	Carbon (approx.)	Moisture / quality	Where found in India	Main use
Peat	Under about **40%**	Highest moisture; first stage of coal; burns like wet wood	Limited swamp / hill pockets (not a major commercial Indian coal)	Local fuel only
Lignite (brown coal)	About **40–55%**	High moisture; low grade; can ignite spontaneously	**Neyveli (Tamil Nadu)**; **Palana / Barsingsar (Rajasthan)**; **Panandhro (Gujarat)**; also J&K / coastal Tertiary notes	Pit-head **thermal power**
Bituminous	About **60–80%**	Most common Indian coal; dense black; high heat	Gondwana valleys — Jharkhand, West Bengal, Odisha, Chhattisgarh, Madhya Pradesh, Telangana, Maharashtra	Power + coke (if coking grade)
Anthracite	About **80–95%**	Hardest; lowest moisture; best quality; burns with little smoke	**Very little** in India — small pockets in **Jammu & Kashmir**	High-efficiency fuel

- About **80%** of India's coal deposits are **bituminous** and of **non-coking** grade (NCERT line).
- **Coking coal** makes metallurgical **coke** for blast furnaces. **Jharia (Jharkhand)** is the main coking field.
- **Non-coking** bituminous coal feeds **thermal power** plants.
- About **four-fifths** of Indian coal production is bituminous **non-coking**.
- **Flyash** is the fine residue of coal-fired thermal plants.

Major Gondwana coalfields

Coalfield	State	Valley / note
Raniganj	**West Bengal**	Damodar; oldest Indian field; first commercial mining **1774**

Coalfield	State	Valley / note
Jharia	**Jharkhand**	Damodar; **coking**; known for mine fires; largest field in many notes
Bokaro / Ramgarh / Karanpura / Giridih	Jharkhand	Damodar family
Rajmahal	Jharkhand	Eastern Jharkhand
Talcher	**Odisha**	**Mahanadi** (महानदी) valley
Ib Valley	Odisha	Mahanadi family
Korba / Mand-Raigarh	Chhattisgarh	Thermal towns
Singrauli	**Madhya Pradesh–Uttar Pradesh**	**Son** valley; **not** northern Chhattisgarh
Sohagpur	Madhya Pradesh	Son family
Singareni	Telangana	**Godavari** (गोदावरी) valley
Wardha / Kamptee / Umrer	Maharashtra	Godavari-Wardha
Coalfield	Valley	
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Talcher	**Mahanadi**	
Karanpura	**Damodar** (दामोदर)	
Singrauli	**Son**	
Singareni	**Godavari**	

Tertiary and lignite centres

Centre	State	Type
Makum / Naharkatiya belt	Assam	Tertiary coal
Maolong	**Meghalaya**	Tertiary (**not Arunachal**)
Namchik–Namphuk	**Arunachal Pradesh** (Changlang)	Tertiary
Neyveli	Tamil Nadu	**Lignite** (main Indian lignite)
Palana / Barsingsar	Rajasthan	Lignite
Panandhro	Gujarat	Lignite

- Lignite is **not** Gondwana bituminous coal.

Reserves, production and institutions

- NTPC **Singrauli** is among NTPC's oldest coal units and sits on the **Madhya Pradesh–Uttar Pradesh** belt.
- **Vindhyachal** (विंध्याचल) in the Singrauli belt (Madhya Pradesh) is among India's largest thermal complexes.
- Commercial chronology starts at Raniganj (**1774**).
- **NCDC** followed in **1956**.
- Coking and non-coking coal were **nationalised in 1971–73**.
- **Coal India Ltd** is the later umbrella company.
- Proved coal-reserve order (Indian Minerals Yearbook, **1 Apr 2021**): **Jharkhand > Odisha > Chhattisgarh > West Bengal > Madhya Pradesh**.
- Coal production order (**2020–21 / 2021–22**): often **Chhattisgarh > Odisha > Madhya Pradesh > Jharkhand**. Reserve order ≠ production order.
- **CBM** is methane from coal seams.
- **Shale gas** is mostly methane from fine shale.
- India has shale potential in Cambay, KG, Cauvery, Damodar, and related basins.
- **Coal Controller's Organisation** keeps coal–lignite statistics, monitors captive blocks, and hears objections under coal-bearing land acquisition law — it does **not** guarantee end-user delivery timelines.

World coalfields (by country)

Field	Country
Appalachian	**USA**
Lancashire	**England**
Ruhr	**Germany**
Kuzbass	**Russia**
Donetsk	**Ukraine**

Inline PYQ — UPPCS Prelims 2019, Q81

Match List-I with List-II and select the correct answer using the codes given below the lists.

List-I (Coal Fields)	List-II (Located)
A. Talcher	1. Damodar Valley
B. Karanpura	2. Son Valley
C. Singrauli	3. Godavari Valley
D. Singareni	4. Mahanadi Valley

> Row order is **not** the answer — use the option codes.

A. 4 1 2 3

B. 3 2 1 4

C. 2 4 3 1

D. 1 3 2 4

Show answer

Ans: A — Talcher–Mahanadi; Karanpura–Damodar; Singrauli–Son; Singareni–Godavari.

N.4 Petroleum, gas, refineries

Petroleum is a **non-renewable** mineral oil found in **sedimentary basins**, not in the Chota Nagpur crystalline core.

Fuel	Main composition	Teaching line
Natural gas / **CNG**	Mainly **methane**	Piped / compressed methane fuel
LPG	**Propane–butane** mix	Not the same as CNG
Petroleum	Mixture of hydrocarbons	Black Revolution = petroleum; Yellow = oilseeds

Indian fields and basins

Field / basin	Note
Digboi	Assam — oldest Indian refinery town / onshore oil
Naharkatiya / Moran / Rudrasagar	Assam–Arakan onshore
Mumbai High	Offshore Arabian Sea; India's giant offshore field
Bassein / Tapti (ताप्ती)	Western offshore gas–oil
Ankleshwar / Cambay / Kalol / Mehsana	Gujarat onshore
KG basin (including KG-D6)	East-coast gas and oil
Ravva	Krishna–Godavari offshore field
Cauvery basin	Tamil Nadu–Puducherry coast
Barmer (**Mangala** and related fields)	Rajasthan onshore

- **Coal-bed methane (CBM)** is gas from coal seams, especially in the Damodar belt.

- The **Hazira–Bijaipur–Jagdishpur (HBJ)** pipeline is the classic gas grid that carried western gas toward north-central markets.

World traps that leak into the Indian paper

- **Pegu Yoma (Bago Yoma, Myanmar)** is **mineral oil**.
- **Kashagan** is in **Kazakhstan**, not Kuwait.
- A 2024-style world set: Iran–Haft Kel, Saudi–Dhahran, Iraq–Zubair are the correct country pairs in that paper.

Refineries

Refineries may sit near **ports** (import crude), near **fields**, or near **markets**. **Mathura** is a market-location refinery for North India. Gujarat concentrates refining because of ports, western crude, and product demand.

Refinery	State
Nayara (Vadinar / Jamnagar belt)	Gujarat
Jamnagar	Gujarat
Koyali (Vadodara)	Gujarat
Manali	Tamil Nadu
Kochi	Kerala
Mangaluru (MRPL)	Karnataka
Numaligarh	Assam
Digboi / Guwahati / Bongaigaon	Assam
Tatipaka	Andhra Pradesh
Visakhapatnam	Andhra Pradesh
Mathura	**Uttar Pradesh**
Barauni	Bihar
Panipat	Haryana
Bhatinda	Punjab
Paradip	Odisha
Bina	Madhya Pradesh
Haldia	West Bengal

Inline PYQ — UPPCS Prelims 2025, Q14

Match List-I with List-II and choose the correct answer using the code given below the lists.

List-I (Refinery)	List-II (State)
A. Nayara	1. Assam
B. Manali	2. Gujarat
C. Numaligarh	3. Andhra Pradesh
D. Tatipaka	4. Tamil Nadu

> Row order is **not** the answer — use the option codes.

A. 4 2 3 1

B. 2 4 1 3

C. 2 4 3 1

D. 4 2 1 3

Show answer

Ans: B — Nayara–Gujarat; Manali–Tamil Nadu; Numaligarh–Assam; Tatipaka–Andhra Pradesh.

N.5 Power: thermal, hydro, nuclear, renewables

Type	Sources	Teaching line
Conventional	Coal, petroleum, natural gas, **nuclear**	Long-used commercial fuels; mostly exhaustible
Non-conventional	Solar, wind, tidal, biogas, geothermal	Cleaner / renewable set in school keys
Stored solar?	Wind, biomass	Nuclear and geothermal are **not** stored solar

Thermal

- A thermal plant burns coal, gas or oil to make steam that turns turbines.
- Coal thermal still dominates Indian electricity.
- **Pit-head** plants sit on coalfields to cut transport cost (Korba, Singrauli, Talcher).
- Thermal plants produce **flyash** and air pollution. Solar does not add that atmospheric pollution in school keys.

Hydroelectricity

Hydel needs a **head** (drop) and a **flow**. Storage dams hold monsoon water; run-of-river plants use live flow.

- The **Himalaya** (हिमालय) holds huge hydel potential.

- Eastern Uttar Pradesh and Bihar have less Himalayan hydel of their own because the high ranges immediately to their north largely lie in **Nepal**.
- **Shivanasamudra** on the Cauvery in Karnataka is a classic hydel site, not a nuclear site.

Hydel name	Note
Shivanasamudra	pair with Kalpakkam
Bhakra	Satluj (multipurpose)
Tehri (टिहरी)	Bhagirathi (भागीरथी)
Hirakud	Mahanadi
Koyna	Maharashtra
Sharavathi / Linganamakki	Karnataka western-ghat hydel
Nagarjuna (नागार्जुन) Sagar (नागार्जुन सागर) / Srisailam	Krishna
Idukki	Kerala

Nuclear plants

Nuclear plants need a large water body for cooling and a secure site. Fuel is **uranium** (and thorium in the long Indian programme), not sunshine.

Plant (परमाणु संयंत्र)	State (राज्य)	Key Facts (मुख्य तथ्य)
Tarapur (तारापुर)	Maharashtra (महाराष्ट्र)	First commercial nuclear plant in India (1969) / भारत का पहला परमाणु संयंत्र
Kakrapar (काकरापार)	Gujarat (गुजरात)	Tapi River / तापी नदी तट
Kudankulam (कुडनकुलम)	Tamil Nadu (तमिलनाडु)	Largest capacity, Russian VVER technology / सबसे बड़ा संयंत्र
Kaiga (कैगा)	Karnataka (कर्नाटक)	Kali River basin / काली नदी बेसिन
Rawatbhata / RAPP (रावतभाटा)	Rajasthan (राजस्थान)	Near Kota, Rana Pratap Sagar / राणा प्रताप सागर
Kalpakkam / MAPS (कल्पक्कम)	Tamil Nadu (तमिलनाडु)	Madras Atomic Power Station / फास्ट ब्रीडर रिएक्टर
Narora (नरोरा)	Uttar Pradesh (उत्तर प्रदेश)	Bulandshahr / बुलंदशहर (गंगा नदी तट, UP का एकमात्र संयंत्र)

- **शिवणसमुद्र (Shivanasamudra)** कर्नाटक में कावेरी नदी पर स्थित **जलविद्युत परियोजना (Hydel project)** है, परमाणु ऊर्जा संयंत्र नहीं।
- **कल्पक्कम (Kalpakkam)** तमिलनाडु में स्थित **परमाणु ऊर्जा संयंत्र (Nuclear plant)** है।
- Nuclear energy (परमाणु ऊर्जा) is **not** biomass (बायोमास नहीं है).
- Nuclear energy and geothermal energy (भू-तापीय ऊर्जा) are **not** ultimately derived from the Sun (सूर्य से व्युत्पन्न नहीं हैं).

- Wind, biomass and most hydel are solar-linked in school energy-cycle keys.
- Coal is a fossil fuel, not “biomass energy (बायोमास ऊर्जा)”. **Gobar gas** and **fuelwood** are biomass.

Inline PYQ — UPPCS Prelims 2025, Q106

Match List-I with List-II and choose the correct answer using the code given below the lists.

List-I (Nuclear Power Plant)	List-II (State)
A. Kudankulam	1. Karnataka
B. Kakrapar	2. Tamil Nadu
C. Kaiga	3. Rajasthan
D. Rawatbhata	4. Gujarat

> Row order is **not** the answer — use the option codes.

A. 4 2 1 3

B. 2 4 3 1

C. 2 4 1 3

D. 4 2 3 1

Show answer

Ans: C — Kudankulam–Tamil Nadu; Kakrapar–Gujarat; Kaiga–Karnataka; Rawatbhata–Rajasthan.

Solar

- Solar electricity is produced mainly through **photovoltaic cells**.
- Among coal, petroleum, nuclear and solar options, **solar** is the clean atmospheric-pollution answer in school keys.
- **Diu** was the first Union Territory framed as running fully on solar.
- **Cochin** is framed as India’s first solar-powered airport.
- **Mirzapur** (मिर्ज़ापुर) in Uttar Pradesh has a major solar project linked with **France** (**2018**).

International Solar Alliance

- India and France launched the ISA in **2015**.
- The ISA Secretariat is at **Gurugram**, not New Delhi or Paris.
- Membership is far beyond any outdated **“38 members”** option.
- **One Sun One World One Grid (OSOWOG)** is an ISA flagship idea.

Wind, tidal, bio-energy, geothermal

- Wind leadership is classically **Tamil Nadu** (**Muppandal**), **Gujarat** (Kutch) and **Rajasthan** (Jaisalmer).

- Tamil Nadu has long been the leading wind-power state in textbook geography.
- Among standard tidal options, the **Gulf of Khambhat** ranks ahead of the **Gulf of Kutch**. The Sundarbans (सुंदरबन) is a smaller tidal option.
- **Biomass / gobar gas** uses cattle dung, crop waste and fuelwood. Coal and nuclear are not biomass.
- **Geothermal** sites such as **Puga (Ladakh (लद्दाख))** and **Manikaran (Himachal (हिमाचल) Pradesh)** are heat from the Earth, not stored solar energy.
- **Tattapani** in Chhattisgarh is another geothermal site.

N.6 Industry, Regions, Corridors

Manufacturing converts raw (रॉ) materials into more valuable goods. Location is decided by a bundle of factors, not one trick alone.

Location factors

- **Raw material** pulls weight-losing industries. Steel and cement sit near bulky ores; cotton fibre can travel farther than iron ore.
- **Power** must be continuous for mills, smelters and refineries.
- **Labour** matters for textiles and engineering, both cheap hands and skilled workers.
- **Market** pulls consumer industries toward cities.
- **Transport** by rail, road and ports cuts cost and also brings imported machines.
- **Capital and banking** help mills start and expand.
- **Government policy** sited public-sector plants and now shapes SEZs and corridors.

Agro-based industries use farm products such as cotton, jute, sugar, rubber and tea. Mineral-based industries use ores and fuels such as iron and steel, cement, aluminium and petrochemicals.

Cotton textile industry

Cotton textile is India’s oldest large-scale industry and still a major employer and export earner.

Why mills first clustered in Maharashtra–Gujarat

- Raw cotton is grown on the Deccan (दक्कन) black-soil belt of Maharashtra and Gujarat, so mills sit close to the fibre.
- Humid coastal air reduces yarn breakage during spinning.
- Mumbai’s port helped import machines and export cloth.
- Rail and road links tied mills to cotton districts and inland markets.
- Cheap labour and trading capital (Parsi / Bhatia merchant finance in early years) supported mill growth.
- Power from western hydel systems fed Mumbai mills; Gujarat later used its own hydel and grid supply.

Mumbai vs Ahmedabad trap

- Mumbai grew as the classic large mill centre (often called India’s cottonopolis in older notes).

- Ahmedabad sits in a major cotton-growing region and became a great mill city, but the still treats **Mumbai** as the traditionally **largest** cotton-textile centre.
- So “Ahmedabad is in the cotton belt” can be true while “Ahmedabad is India’s largest mill centre” is false.

Decentralisation after the early cluster

- Congestion, labour unrest and rising costs pushed mills out of Mumbai’s core.
- Irrigation and hydel (for example Mettur / Pykara in the south) helped new cotton areas.
- Important centres now include Ahmedabad, Surat, Vadodara, Coimbatore, Madurai (मदुरै), Indore, Solapur, Nagpur, Kanpur and other inland towns.
- **Coimbatore** is the main Tamil Nadu spinning / hosiery belt and the main southern spinning centre.
- **Kanpur** (कानपुर) is the classic Uttar Pradesh cotton-textile and leather city in older industrial geography.

Other textile belts

- Jute mills cluster on the **Hugli** belt because raw jute from the Ganga (गंगा)–Brahmaputra (ब्रह्मपुत्र) delta (डेल्टा) meets Kolkata’s port and humid processing climate.
- Woollen textiles favour cooler north-west markets and raw-wool areas (Punjab–Haryana–Rajasthan).
- Silk has strong historic belts in Karnataka, Assam, West Bengal and Kashmir / Varanasi (वाराणसी) craft traditions.

Inline PYQ — UPPCS Prelims 2020, Q58

Given below are two statements, one labelled as Assertion (A) and the other as Reason (R):

Assertion (A): Ahmedabad is the largest centre of cotton textile industry in India.

Reason (R): Ahmedabad is located in a major cotton-growing region of India, so it has no problem of raw materials.

A. Both (A) and (R) are true and (R) is the correct explanation of (A)

B. Both (A) and (R) are true but (R) is not the correct explanation of (A)

C. (A) is true but (R) is false

D. (A) is false but (R) is true

Show answer

Ans: D — Assertion is false because Mumbai is traditionally treated as the largest centre. Reason is true because Ahmedabad does lie in a major cotton region.

Rubber industry

- Natural rubber is an **agro-based** industry tied to plantation geography.
- **Kerala** dominates India’s natural-rubber area and production.
- Tamil Nadu and other humid belts add smaller shares.
- Tyre and product plants often sit near **auto markets** and ports, not only beside every plantation.

- Synthetic rubber follows the **petrochemical** chain, not Kerala gardens.

Jute, sugar, cement, petrochemicals, automobiles

- **Jute** mills cluster on the **Hugli** belt because raw jute from the Ganga–Brahmaputra delta meets Kolkata's port and humid processing climate.
- **Sugar** is a **weight-losing** industry, so mills sit in the cane belt.
- Uttar Pradesh leads in **quantity**.
- Maharashtra often leads in **recovery / productivity**.
- **Cement** follows bulky **limestone**. Plants sit near quarries, then send bags to markets.
- **Petrochemicals** follow refineries and gas (Gujarat, Mumbai–Pune (पुणे), Panipat).
- **Automobile** clusters include Pune–Mumbai, Chennai–Bengaluru–Hosur, NCR (Gurugram), and Gujarat's newer auto belt.

Iron and steel

Iron and steel is a **basic / key** industry because it feeds machine tools, transport, construction and defence.

Steel is a **weight-losing** industry when both ore and coking coal are bulky, so plants grew on the Chotanagpur–Odisha–Chhattisgarh mineral belt.

Plant	State
Bhilai	Chhattisgarh
Rourkela	Odisha
Bokaro	Jharkhand
Durgapur / Burnpur	West Bengal
Visakhapatnam	Andhra Pradesh
Salem	Tamil Nadu
Bhadravati / Vijayanagar	Karnataka
Jamshedpur	Jharkhand (Tata (तत) pioneer)

- The Chotanagpur–Odisha–Chhattisgarh belt has the densest steel cluster because high-grade iron ore, coking coal, limestone/dolomite flux, water and labour sit close together.
- Coastal plants such as Visakhapatnam also use port logistics for ore and markets.
- Aluminium smelting is power-hungry, so plants sit near cheap electricity as well as bauxite.
- The **BALCO** plant is at **Korba**.
- The **HINDALCO** plant at **Renukoot** is in **Uttar Pradesh**.
- Uttar Pradesh is **not** a Bhilai-class integrated steel state.

Industrial regions

Region	Core character
Mumbai–Pune	Cotton textiles, engineering, chemicals, automobiles
Hugli	Jute, engineering, port industries
Bangalore–Tamil Nadu belt	Diversified engineering, autos, textiles; Bengaluru IT + Chennai + Coimbatore
Ahmedabad–Vadodara–Surat (Gujarat)	Textiles, petrochemicals, engineering
Chotanagpur	Mineral-based heavy industry — India's Ruhr industrial extension
Visakhapatnam–Guntur	Coastal steel, shipbuilding, engineering
Gurgaon–Delhi–Meerut	NCR automobiles, electronics; western Uttar Pradesh link
Kollam–Thiruvananthapuram	Kerala coastal-plantation and light industry

World city–industry pairs often tested: **Osaka**–cotton textile, **Detroit**–automobile, **Cuba**–cigar, **St Petersburg**–ship building.

Agencies, SEZs and corridors

Agency	Headquarters
ONGC	New Delhi
MECL	Nagpur
NMDC	Hyderabad
NALCO	Bhubaneswar

A **Special Economic Zone (SEZ)** is an export-oriented enclave with duty and tax incentives. **Kandla** was the historic first SEZ / EPZ.

Corridors cut logistics cost between ports, metros and manufacturing clusters. **Dedicated Freight Corridors** move bulk freight off passenger tracks.

Corridor / zone	Meaning
DMIC	Delhi–Mumbai Industrial Corridor — the must-know industrial corridor
Western DFC	Dedicated freight rail **Dadri–JNPT**
Eastern DFC	Dedicated freight rail **Ludhiana–Dankuni**
CBIC	Chennai–Bengaluru Industrial Corridor
AKIC	Amritsar (अमृतसर)–Kolkata Industrial Corridor

Corridor / zone	Meaning
VCIC	Visakhapatnam–Chennai Industrial Corridor
SEZ	Export-oriented special zone

Inline PYQ — UPPCS Prelims 2025, Q9

Which of the following are correctly matched?

Coalfield	State (as given in options)
1. Maolong	Arunachal Pradesh
2. Raniganj	West Bengal
3. Ramgarh	Jharkhand
4. Talcher	Odisha

A. Only 1, 2 and 3

B. Only 2, 3 and 4

C. Only 1, 3 and 4

D. 1, 2, 3 and 4

Show answer

Ans: B — Only 2, 3 and 4. Maolong is in **Meghalaya**, not Arunachal Pradesh.

N.7 Uttar Pradesh focus

Uttar Pradesh is not a Jharkhand-scale ore state. The **Gangetic plains** are almost empty of metallic ores. The mineral and energy belt is the **southern hard-rock / Vindhyan** fringe, especially **Sonbhadra**.

- **Sonbhadra** is the energy and mineral capital of the state: non-coking coal, limestone, dolomite, thermal plants, and **HINDALCO Renukoot**.
- Singrauli–Sonbhadra coal is **non-coking thermal coal**, not Jharia coking coal.
- **Chunar, Churk and Dalla** are limestone–cement towns.
- **Shankargarh** near Prayagraj supplies **silica sand** to **Firozabad** glass.
- **Sonrai in Lalitpur** is a copper occurrence, not a Khetri-class belt.
- **Obra, Anpara and Rihand** are thermal plants of the Singrauli–Sonbhadra belt.
- **Vindhyachal** is on the Madhya Pradesh side of the same belt.
- **Narora** is Uttar Pradesh's nuclear power station.
- **Mathura** is the state's petroleum refinery (market location for North India).
- **Mirzapur** has a major solar project linked with France (**2018**).

City	Industry tag
Kanpur	Textile, leather
Moradabad	Brass
Aligarh	Padlocks and hardware
Firozabad	Glass
Noida–Ghaziabad	NCR engineering / electronics
Meerut	NCR belt; **sports goods**
Agra	Light industry and tourism, not a mineral capital
Renukoot	Aluminium (HINDALCO)
Chunar–Dalla belt	Cement